

Chemical Reactor Network Modelling For Iron Particle Combustion

Bachelor Thesis Project – 1



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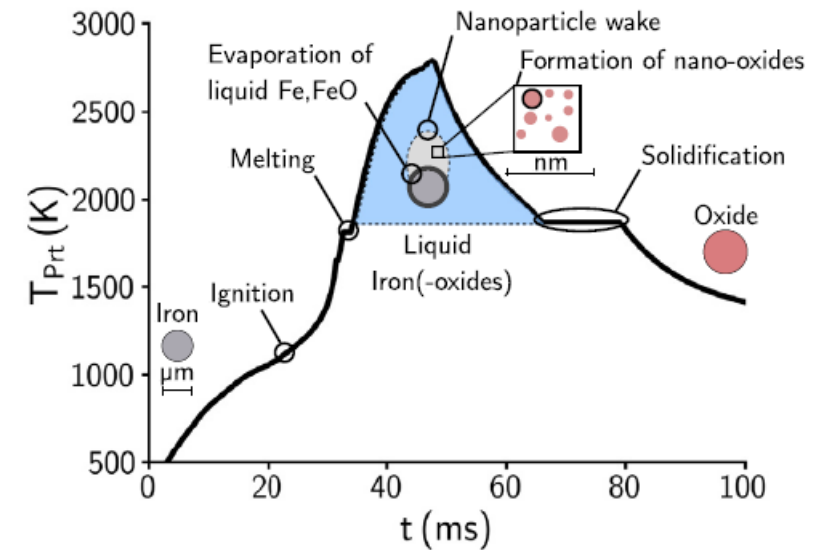
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Introduction

- Metal fuels such as iron powder are promising alternatives for clean energy
- When iron reacts with oxygen, it releases significant heat energy
- Iron combustion can be used for **carbon-free energy systems**
- Iron powder is especially promising due to its practicality

Challenges in modelling Iron Combustion

- Multi-phase system (solid + gas)
- Multiple processes:
 - Heating, melting, oxidation, evaporation
- Oxidation pathway:
 - $\text{Fe} \rightarrow \text{FeO} \rightarrow \text{Fe}_3\text{O}_4 \rightarrow \text{Fe}_2\text{O}_3$
- Strong coupling between heat, mass transfer, and reactions
- **Conclusion:** Direct simulation is complex and computationally expensive



Why Iron Is A Good Fuel

- Non-toxic
- High energy density
- Already produced in large quantities
- Can be recycled
- **Iron cycle:**
 - Iron → Burn → Iron Oxide → Recycle using Hydrogen → Iron again
- This creates a **circular energy system**

Experimental Setup :

- Setup used:
 - Laboratory-scale combustion chamber
 - Methane-assisted iron dust flame
 - Swirling air flow for mixing
- Purpose of study:
 - flame structure
 - Study oxygen distribution
 - Investigate pollutant formation
- Results compared with:
 - CFD simulations and LES data.

Models Used in Study :

- Two oxidation models.
 - **1. FOSK Model**
 - First Order Surface Kinetics
 - oxidation occurs at the particle surface
 - reaction depends on oxygen diffusion
 - **2. Reactive Cooling Model**
 - considers layered oxide structure
 - includes evaporation effects
 - captures detailed particle physics
- These models simulate iron particle conversion during combustion.

What is a Chemical Reactor Network (CRN)?

- A CRN is a combination of simple reactors used to represent a complex combustion system.
- Reactor types :
 - **1. Perfectly Stirred Reactor (PSR)**
- Complete mixing
- Used for recirculation zones
- **2. Plug Flow Reactor (PFR)**
- One-directional (plug) flow
- No axial mixing
- PSR + PFR together simulate real combustion chamber behavior
- *CRN has been successfully applied to iron particle combustion in literature*

Objective of This Work

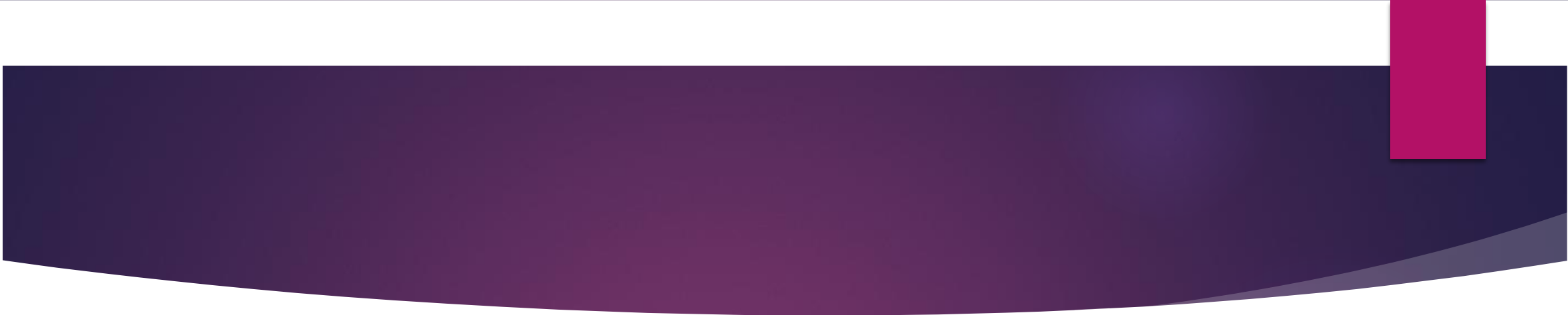
- Develop CRN models for iron combustion
- Advantage:
- Lower cost than CFD
- However, iron combustion is highly complex due to multi-phase interactions.

0D Reactor as a starting point

- A **0D Ideal Gas Reactor** is used
 - Perfect mixing
 - No spatial variation
 - Only time-dependent changes
- **0D reactor :**
 - Basically similar to a **PSR concept (perfect mixing)**
 - A **simplest reactor model**
 - Used as the simplest CRN element to understand combustion behavior

Conclusions :

- **From Literature (Iron CRN):**
- CRN models iron combustion effectively
- Oxygen controls reaction
- Temperature affects NO_x & evaporation.
- Particle size affects evaporation



Thank you